Haokun (Daniel) Xu

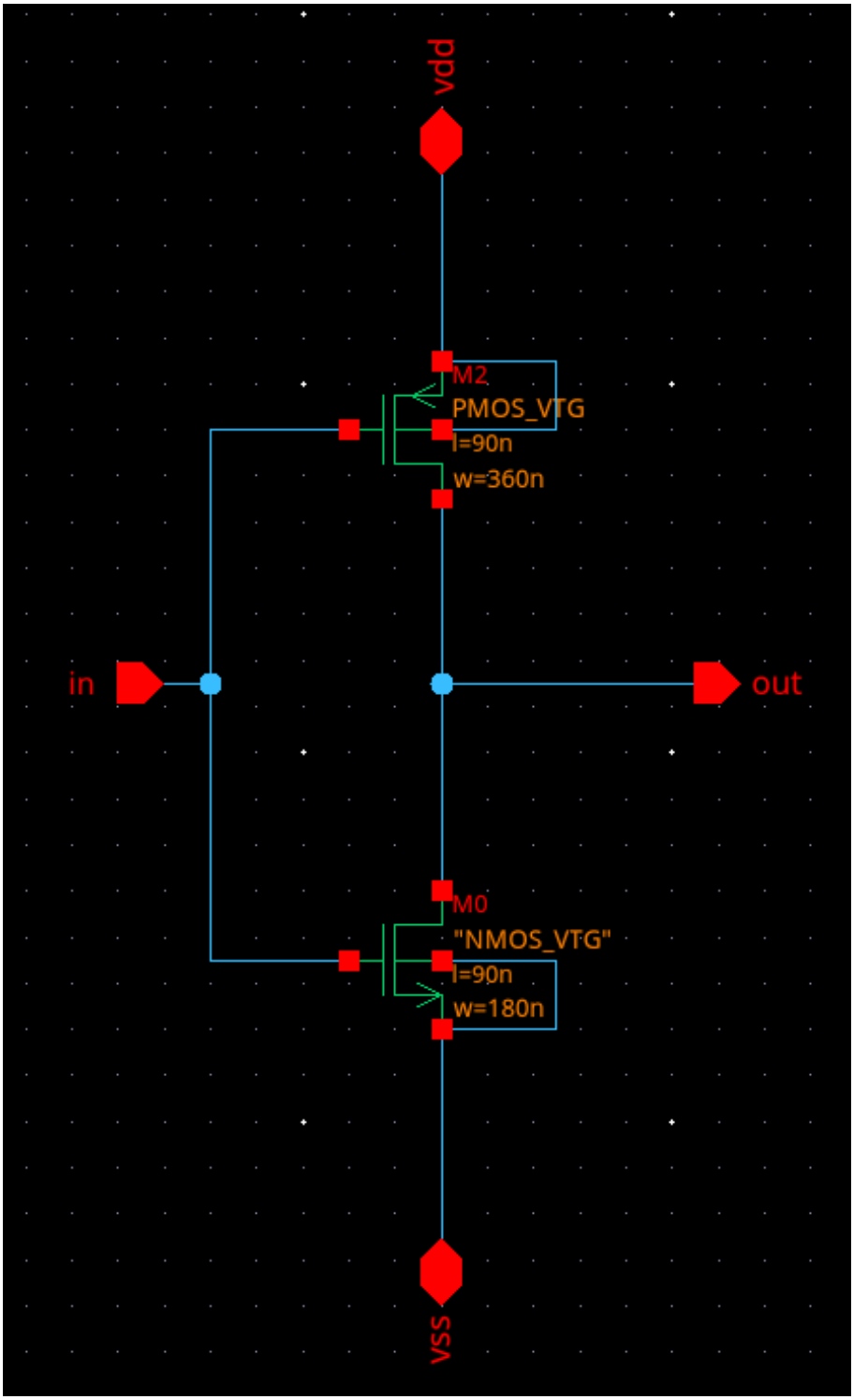
ECE 4385

HW 1

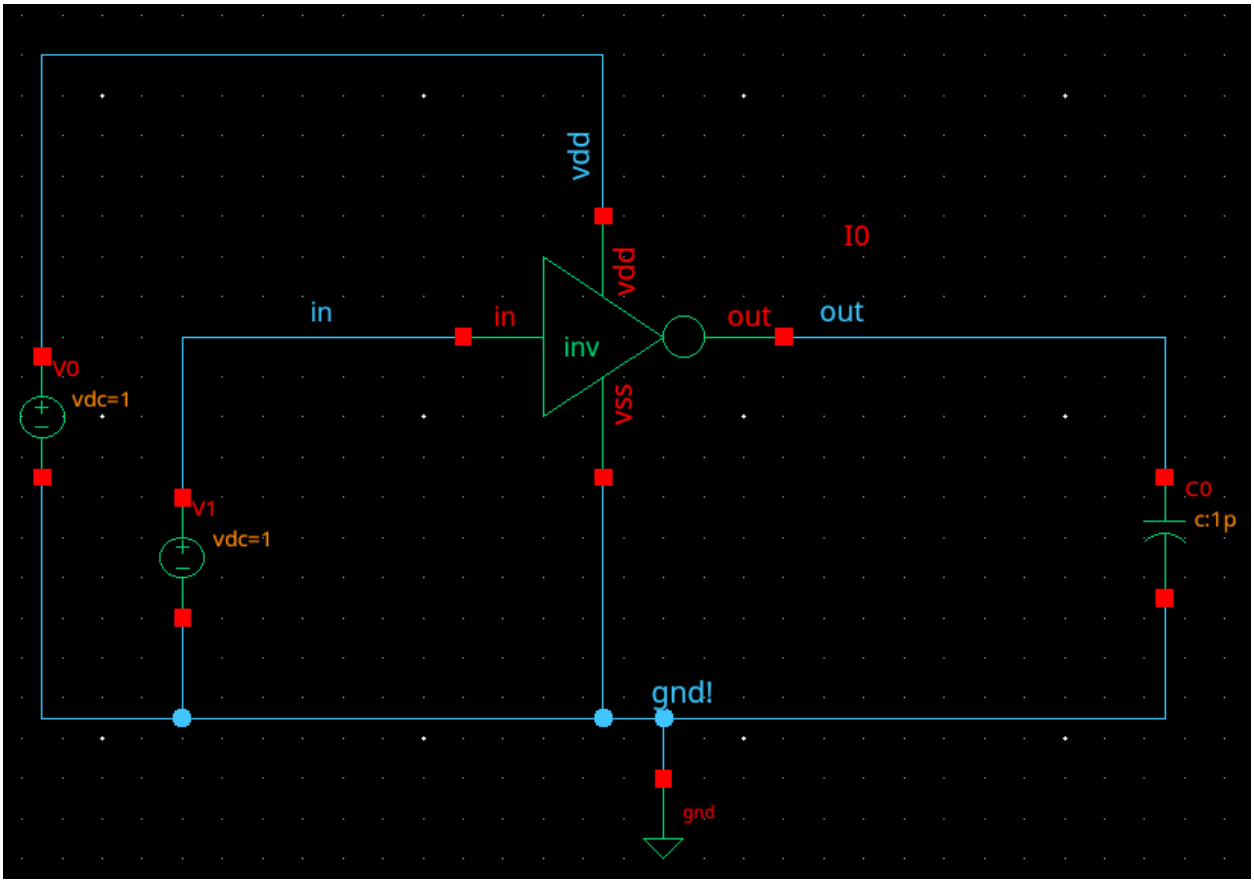
I understand some of these screenshots don’t have my name on them. I couldn’t find a way to show it on the screenshot. Sorry ☹ I have uploaded my files as a zip to show my work (and I have commit history on github if needed)

Inverter

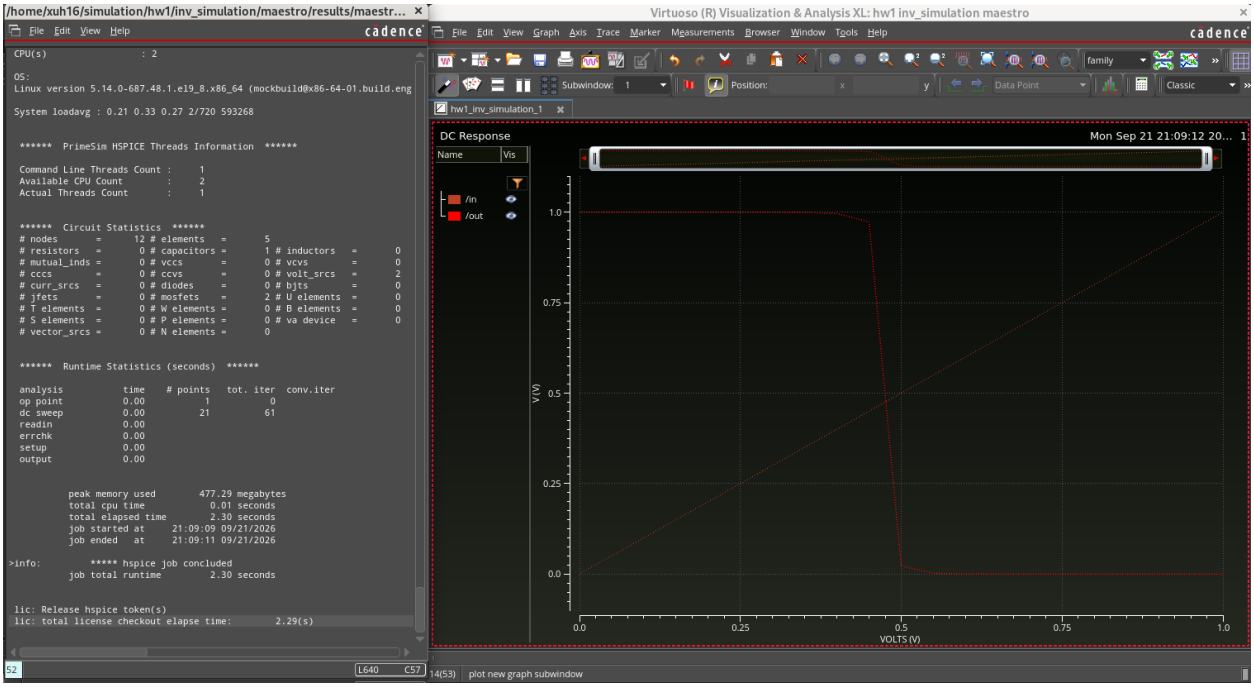
Schematic



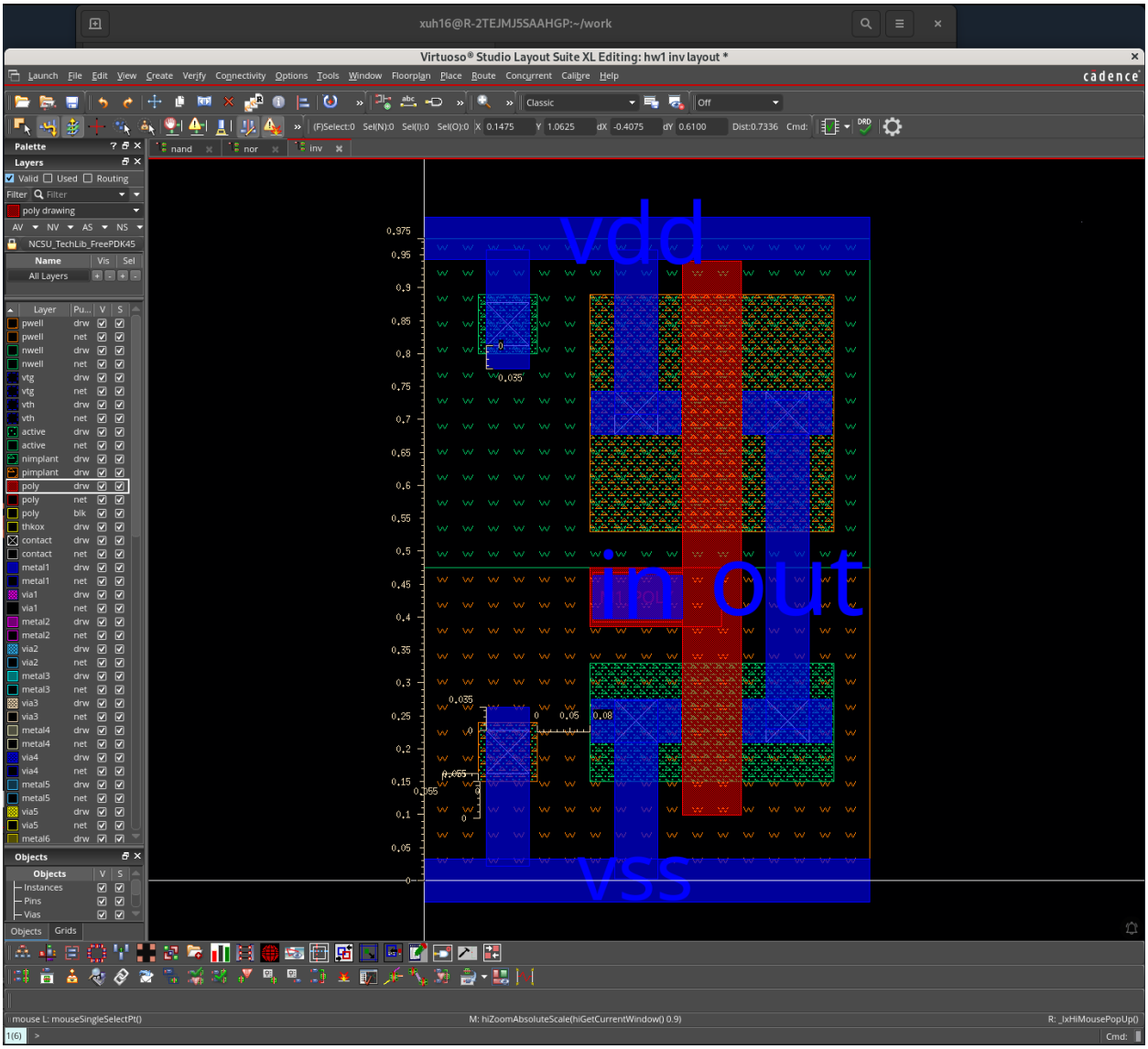
Simulation Schematic



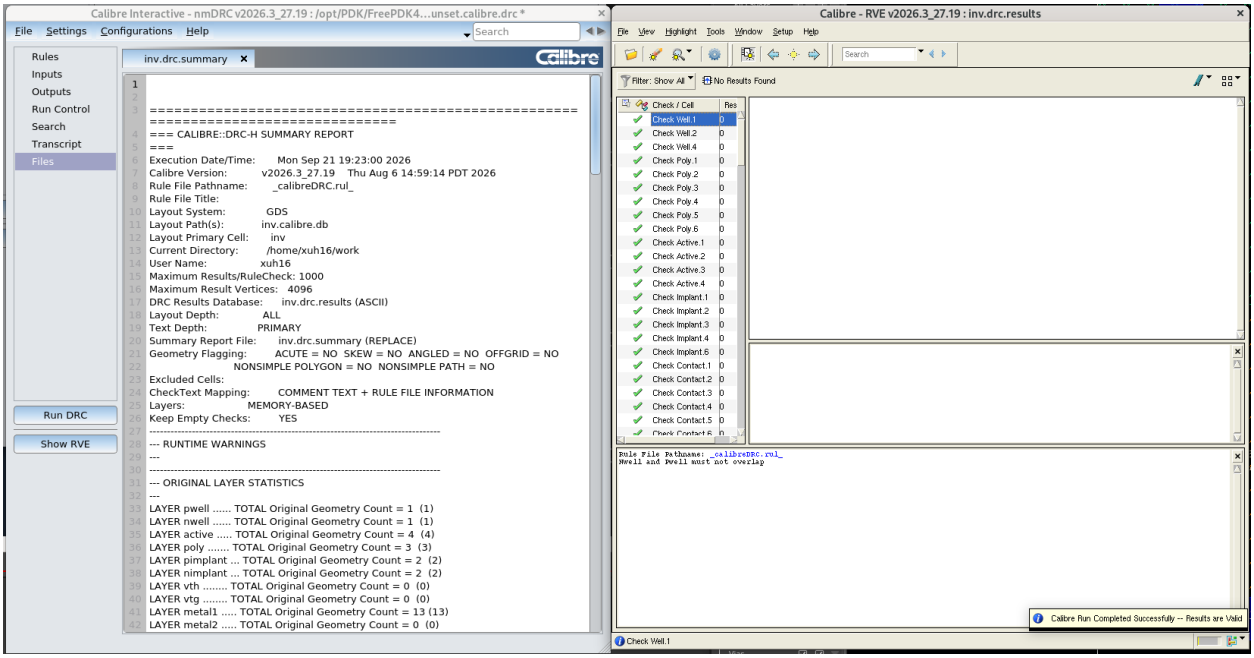
Simulation Results



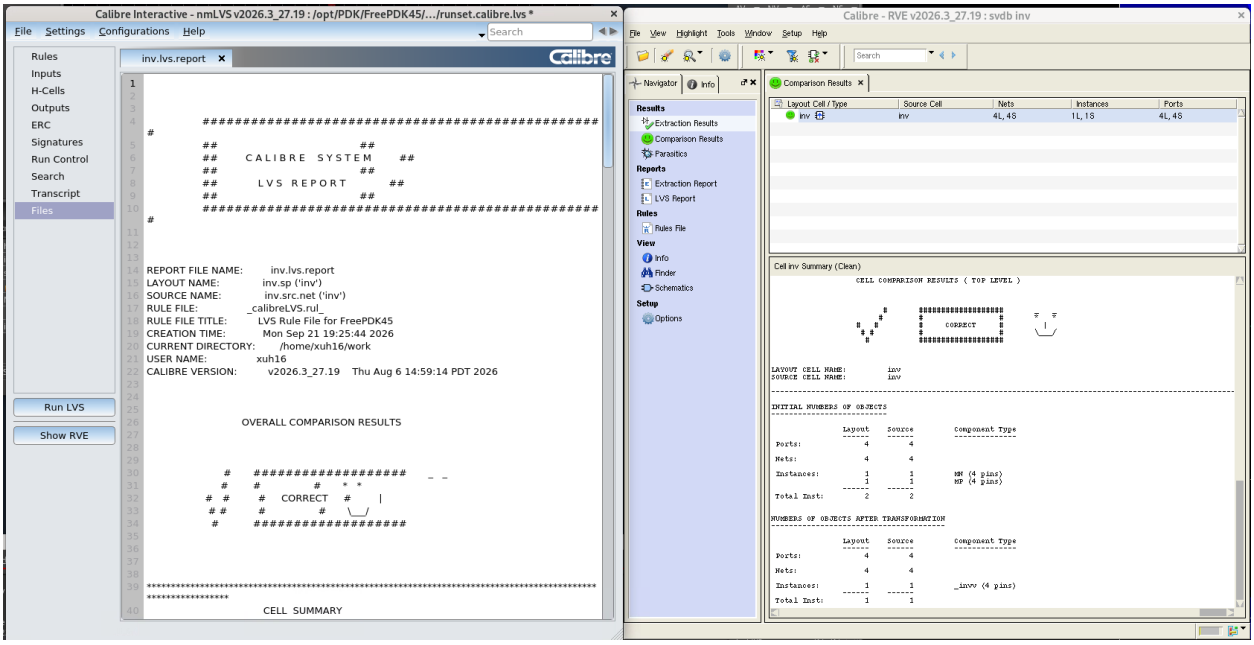
Layout


xuh16@R-2TEJMJ5SAAHGP:~/work
Virtuoso® Studio Layout Suite XL Editing: hw1 inv layout *
vdd
in
out
vss

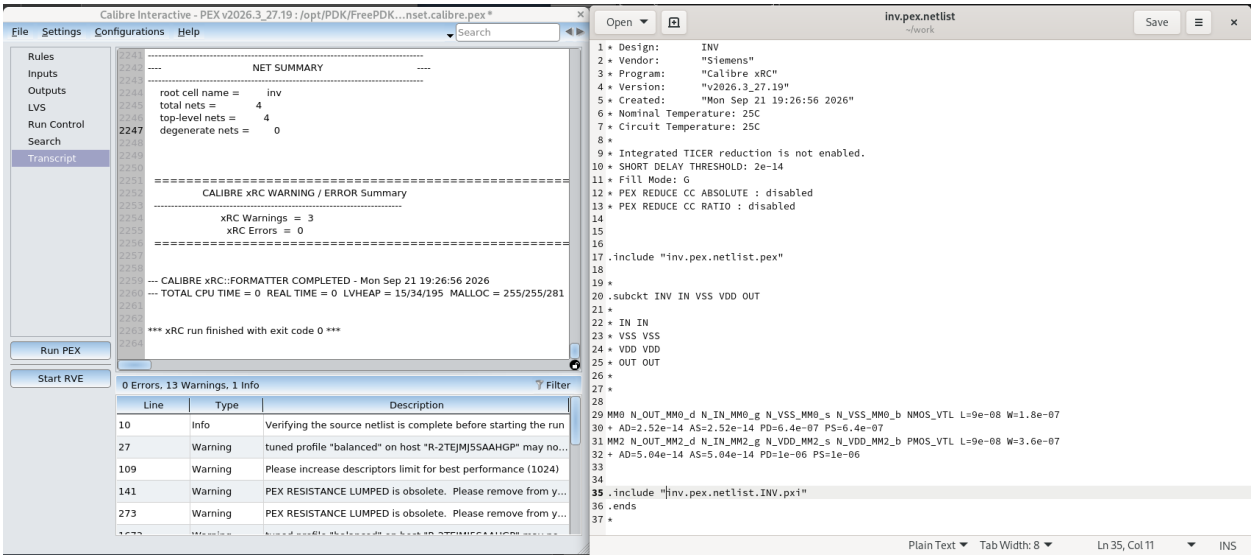
DRC


Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
File Settings Configurations Help
Rules
Inputs
Outputs
Run Control
Search
Transcript
Files
Run DRC
Show RVE
inv.drc.summary
=== CALIBRE::DRC-H SUMMARY REPORT
Execution Date/Time: Mon Sep 21 19:23:00 2026
Calibre Version: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
Rule File Pathname: _calibreDRC.rul_
Rule File Title:
Layout System: GDS
Layout Path(s): inv.calibre.db
Layout Primary Cell: inv
Current Directory: /home/xuh16/work
User Name: xuh16
Maximum Results/RuleCheck: 1000
Maximum Result Vertices: 4096
DRC Results Database: inv.drc.results (ASCII)
Layout Depth: ALL
Text Depth: PRIMARY
Summary Report File: inv.drc.summary (REPLACE)
Geometry Flagging: ACUTE = NO SKEW = NO ANGLED = NO OFFGRID = NO
NONSIMPLE POLYGON = NO NONSIMPLE PATH = NO
Excluded Cells:
CheckText Mapping: COMMENT TEXT + RULE FILE INFORMATION
Layers: MEMORY-BASED
Keep Empty Checks: YES
--- RUNTIME WARNINGS
--- ORIGINAL LAYER STATISTICS
LAYER pwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER nwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER active ..... TOTAL Original Geometry Count = 4 (4)
LAYER poly ..... TOTAL Original Geometry Count = 3 (3)
LAYER pimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER nimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER vth ..... TOTAL Original Geometry Count = 0 (0)
LAYER vtg ..... TOTAL Original Geometry Count = 0 (0)
LAYER metal1 ..... TOTAL Original Geometry Count = 13 (13)
LAYER metal2 ..... TOTAL Original Geometry Count = 0 (0)
Calibre - RVE v2026.3_27.19 : inv.drc.results
File View Highlight Tools Window Setup Help
Filter: Show All
No Results Found
Check / Cell
Check Well.1
Check Well.2
Check Well.4
Check Poly.1
Check Poly.2
Check Poly.3
Check Poly.4
Check Poly.5
Check Poly.6
Check Active.1
Check Active.2
Check Active.3
Check Active.4
Check Implant.1
Check Implant.2
Check Implant.3
Check Implant.4
Check Implant.6
Check Contact.1
Check Contact.2
Check Contact.3
Check Contact.4
Check Contact.5
Rule File Pathname: _calibreDRC.rul_
Nwell and Pwell must not overlap
Calibre Run Completed Successfully -- Results are Valid
Check Well.1

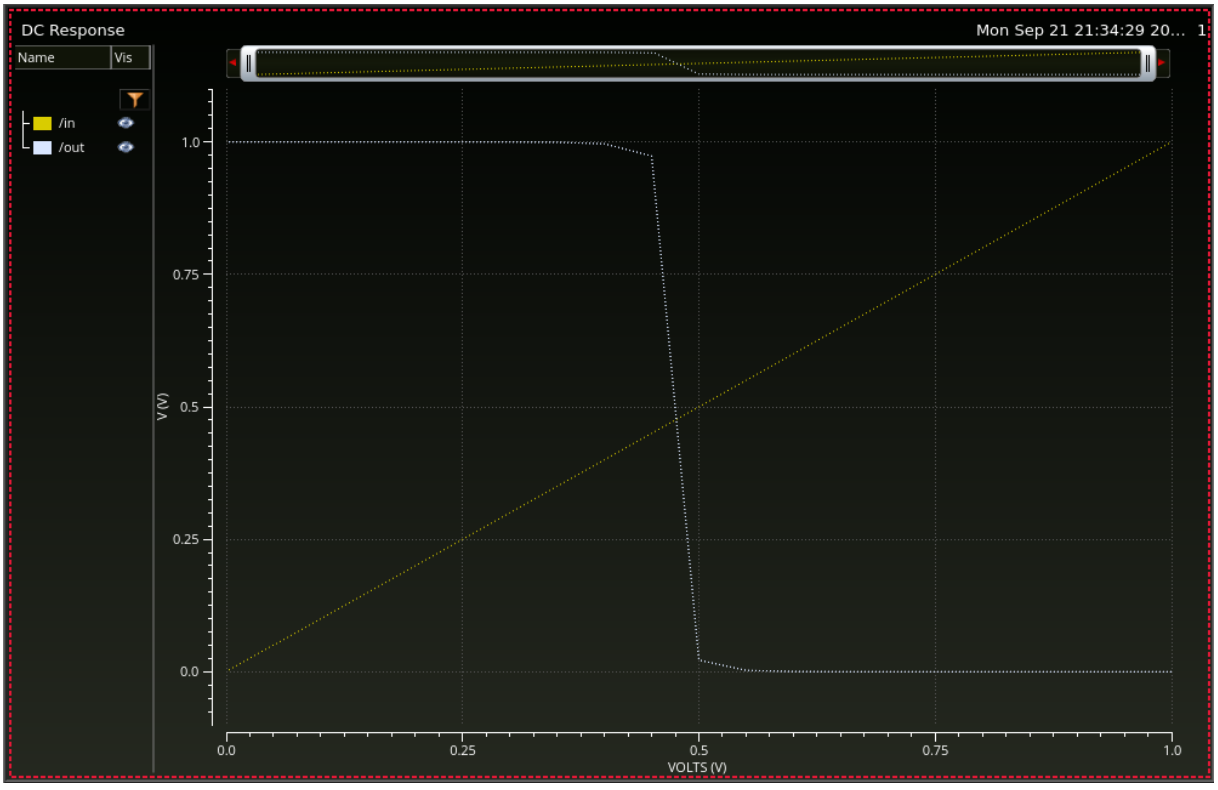
LVS

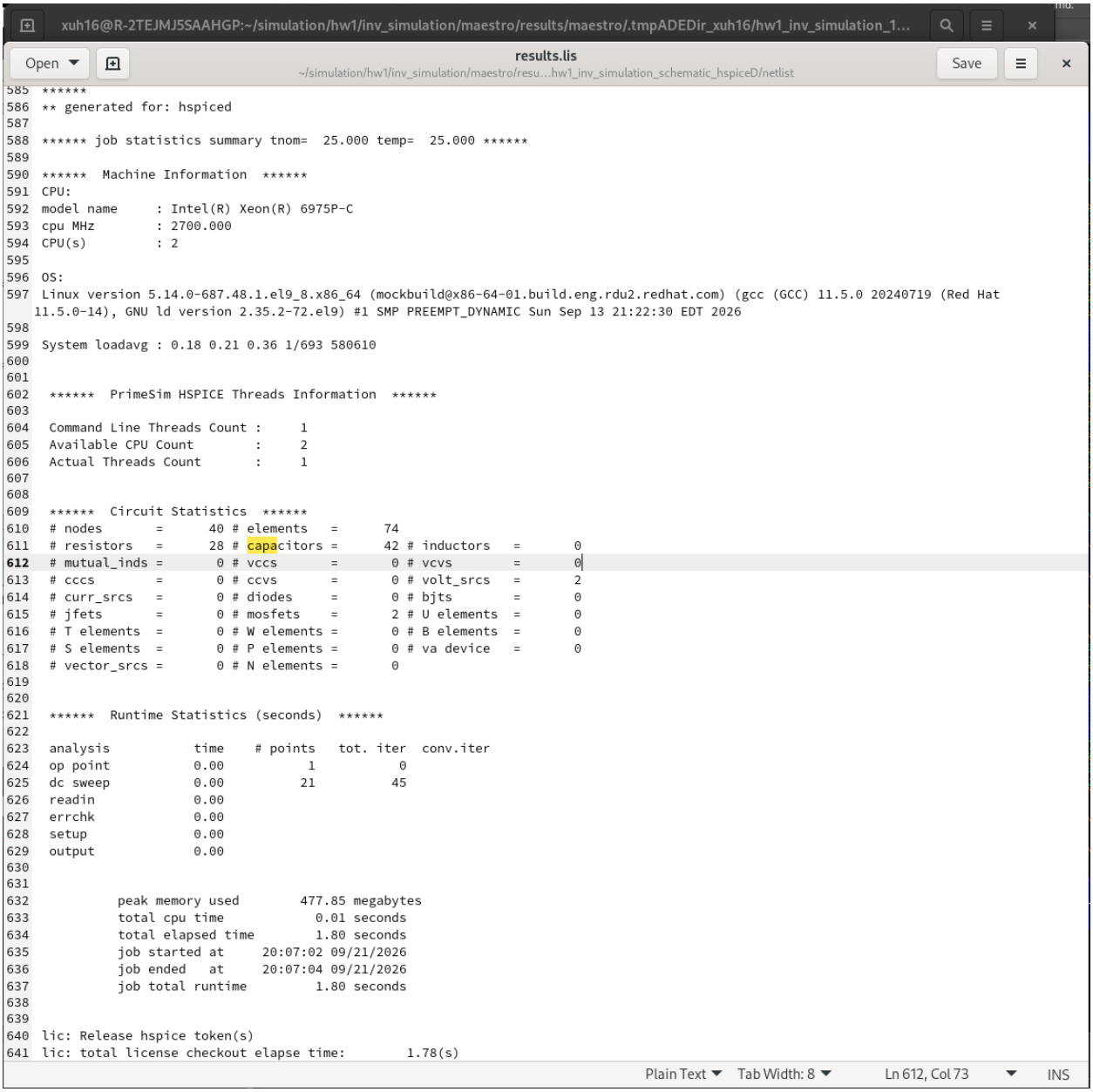

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK45/.../runset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
inv.lvs.report
CALIBRE SYSTEM
LVS REPORT
REPORT FILE NAME: inv.lvs.report
LAYOUT NAME: inv.sp ('inv')
SOURCE NAME: inv.src.net ('inv')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Sep 21 19:25:44 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
CELL SUMMARY
Calibre - RVE v2026.3_27.19 : svdb inv
File View Highlight Tools Window Setup Help
Navigator
Info
Results
Extraction Results
Comparison Results
Parasitics
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
inv
Cell inv Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: inv
SOURCE CELL NAME: inv
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 4 4
Nets: 4 4
Instances: 1 1 NM (4 pins)
1 1 MP (4 pins)
Total Inst: 2 2
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Layout Source Component Type
Ports: 4 4
Nets: 4 4
Instances: 1 1 _invv (4 pins)
Total Inst: 1 1

PEX



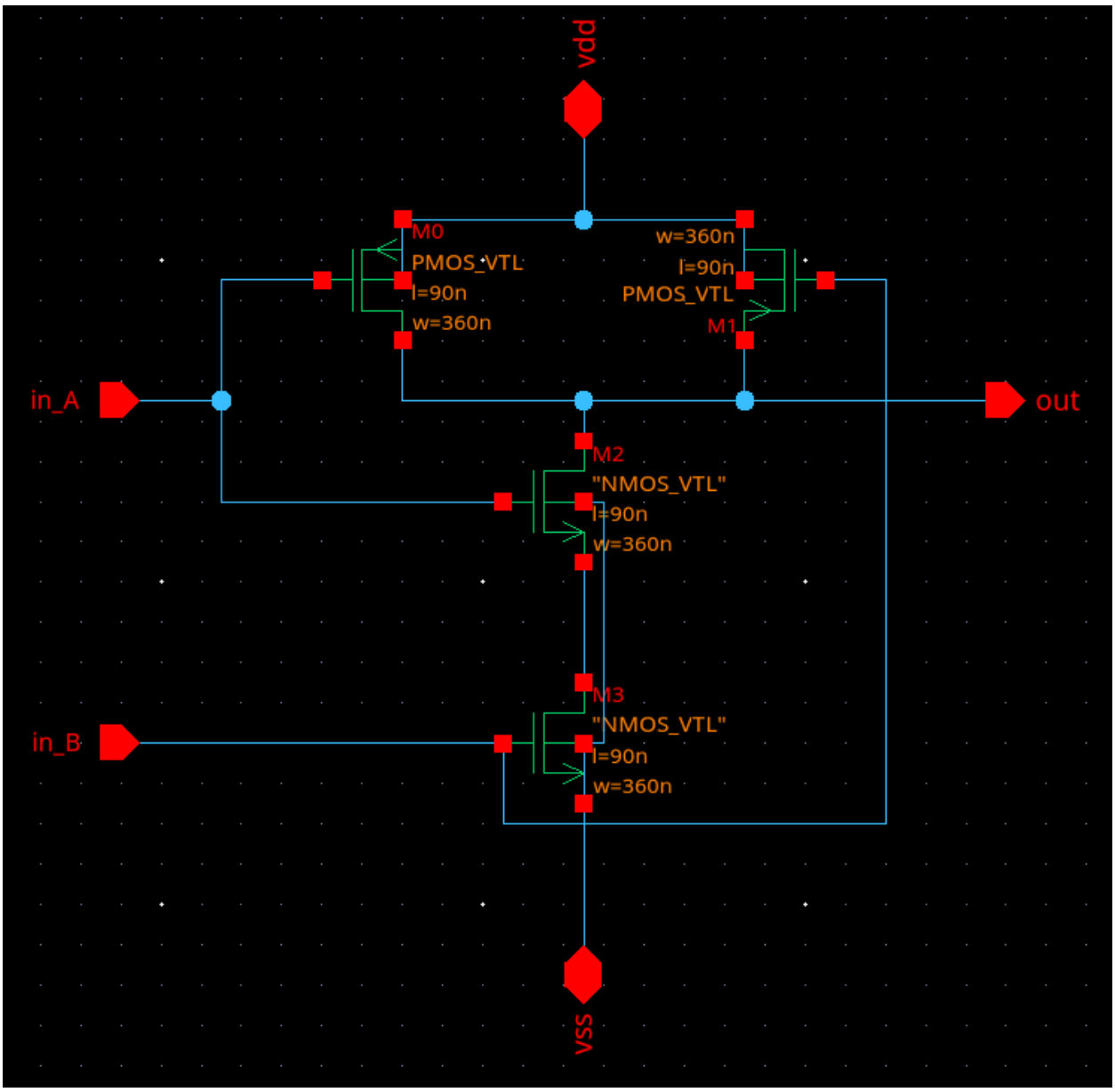
Post-Layout Simulation



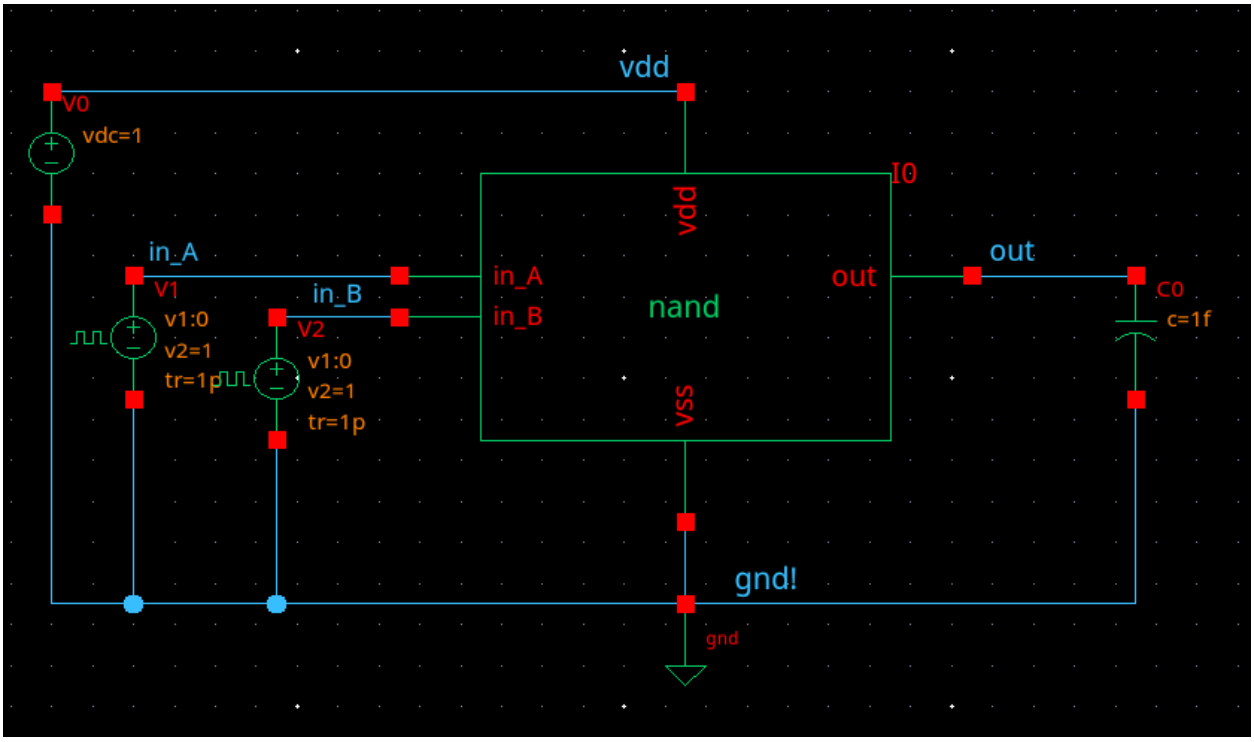

xuh16@R-2TEJMJ5SAAHGP:~/simulation/hw1/inv_simulation/maestro/results/maestro/.tmpADEDir_xuh16/hw1_inv_simulation_1...
Open
results.lis
~/simulation/hw1/inv_simulation/maestro/resu...hw1_inv_simulation_schematic_hspiceD/netlist
Save
****** job statistics summary tnom= 25.000 temp= 25.000 ******
****** Machine Information ******
CPU:
model name : Intel(R) Xeon(R) 6975P-C
cpu MHz : 2700.000
CPU(s) : 2
OS:
Linux version 5.14.0-687.48.1.el9_8.x86_64 (mockbuild@x86-64-01.build.eng.rdu2.redhat.com) (gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-14), GNU ld version 2.35.2-72.el9) #1 SMP PREEMPT_DYNAMIC Sun Sep 13 21:22:30 EDT 2026
System loadavg : 0.18 0.21 0.36 1/693 580610
****** PrimeSim HSPICE Threads Information ******
Command Line Threads Count : 1
Available CPU Count : 2
Actual Threads Count : 1
****** Circuit Statistics ******
nodes = 40 # elements = 74
resistors = 28 # capacitors = 42 # inductors = 0
mutual_inds = 0 # vccs = 0 # vcvs = 0
cccs = 0 # ccvs = 0 # volt_srcs = 2
curr_srcs = 0 # diodes = 0 # bjts = 0
jfets = 0 # mosfets = 2 # U elements = 0
T elements = 0 # W elements = 0 # B elements = 0
S elements = 0 # P elements = 0 # va device = 0
vector_srcs = 0 # N elements = 0
****** Runtime Statistics (seconds) ******
analysis time # points tot. iter conv.iter
op point 0.00 1 0
dc sweep 0.00 21 45
readin 0.00
errchk 0.00
setup 0.00
output 0.00
peak memory used 477.85 megabytes
total cpu time 0.01 seconds
total elapsed time 1.80 seconds
job started at 20:07:02 09/21/2026
job ended at 20:07:04 09/21/2026
job total runtime 1.80 seconds
lic: Release hspice token(s)
lic: total license checkout elapse time: 1.78(s)
Plain Text
Tab Width: 8
Ln 612, Col 73
INS

NAND

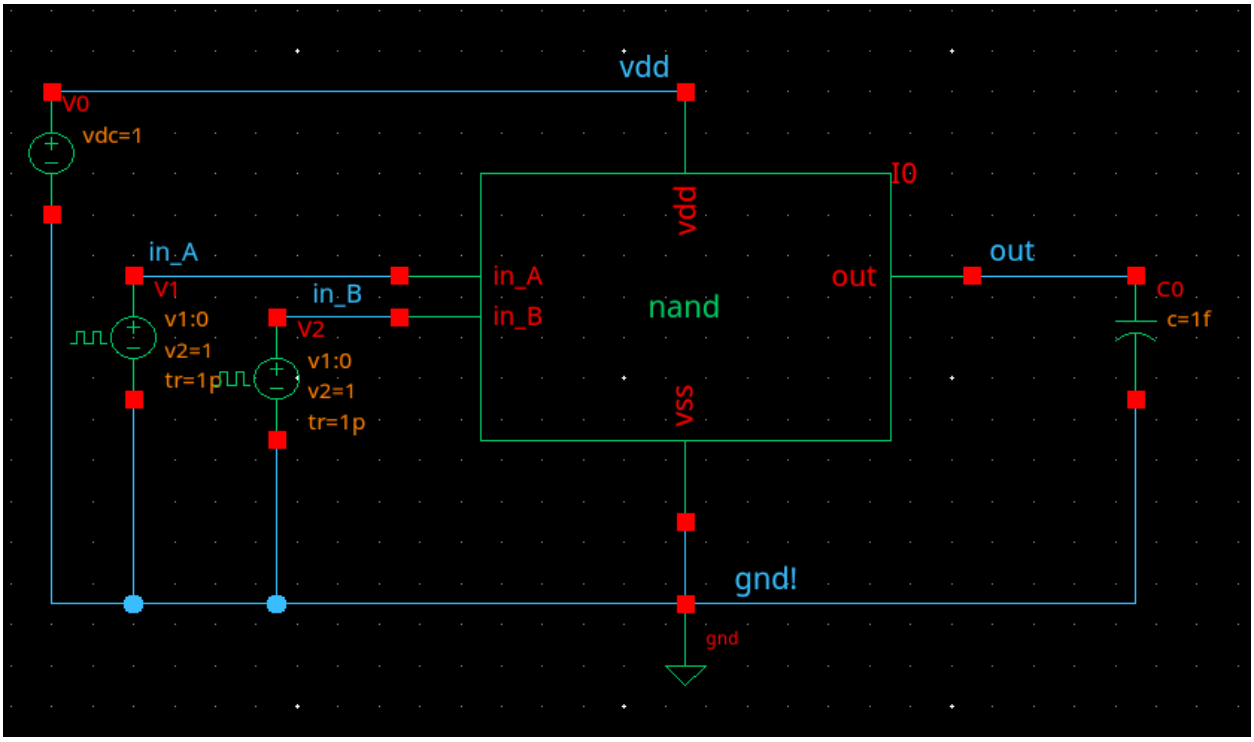
Schematic



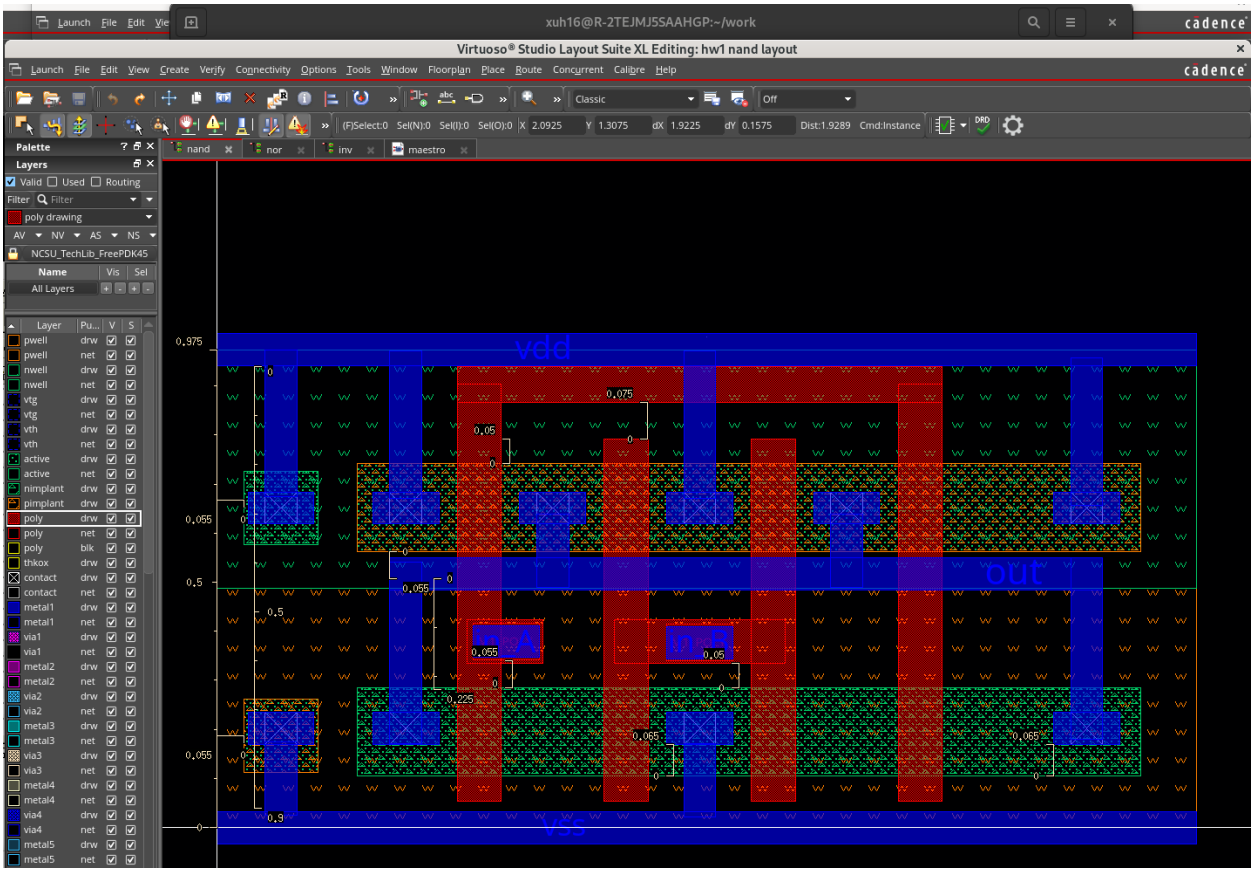
Simulation Schematic



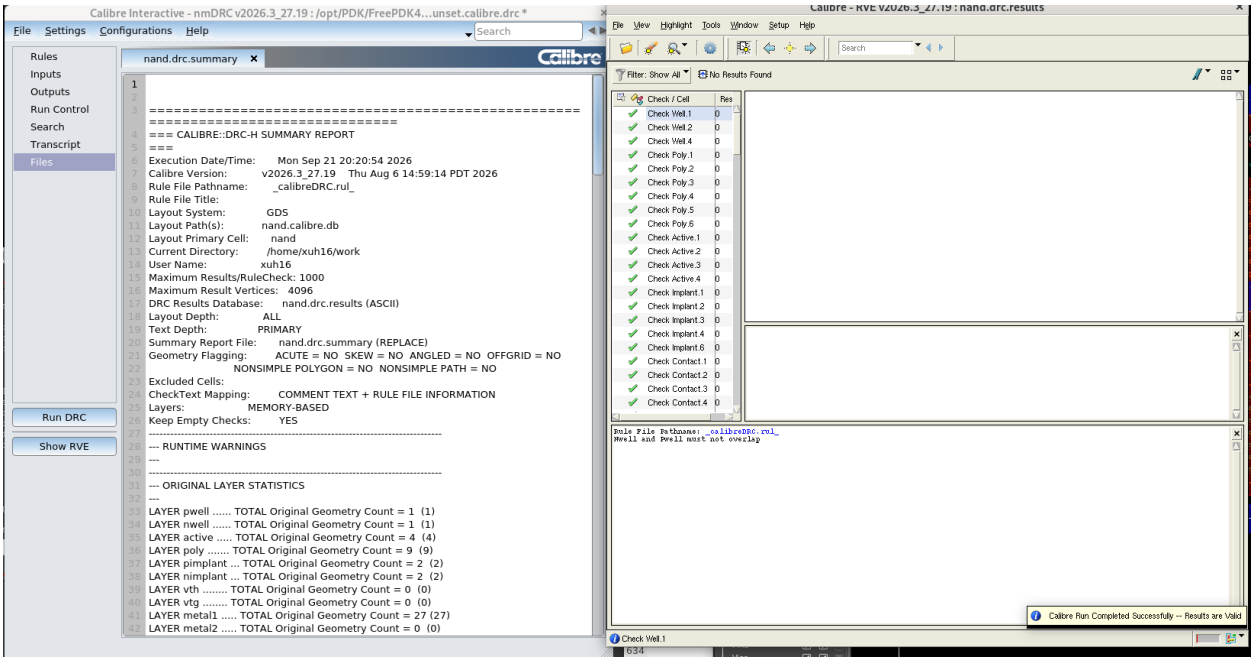
Simulation Results



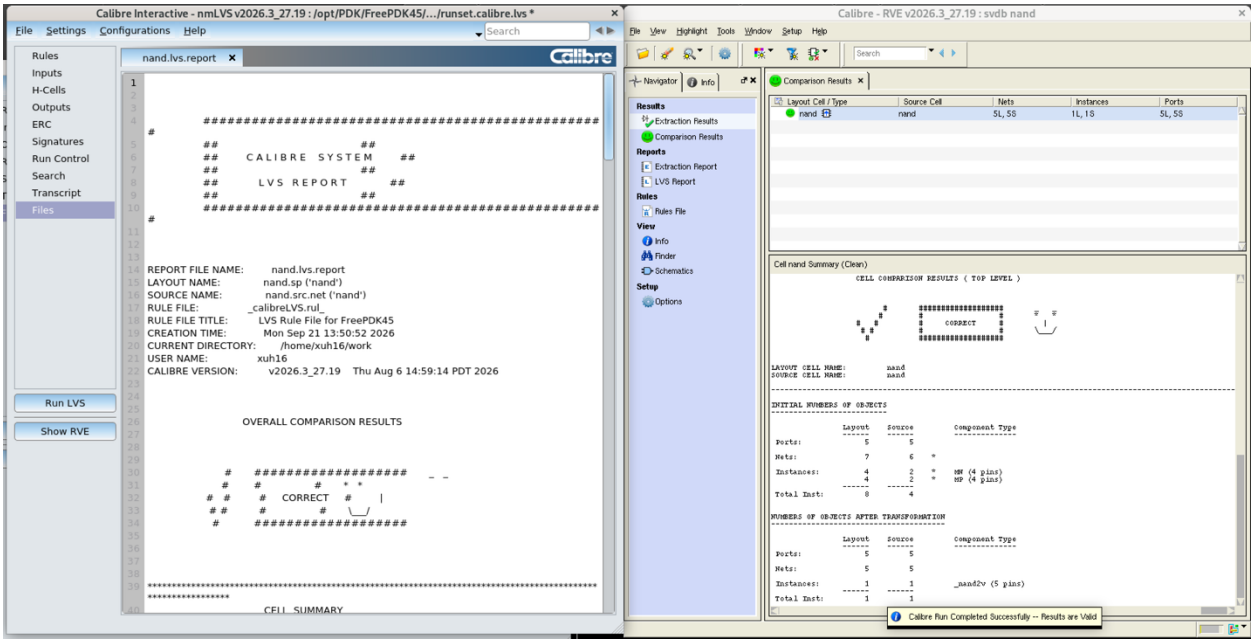
Layout


Virtuoso® Studio Layout Suite XL Editing: hw1 nand layout
vdd
vss
out
inA
inB
0.975
0.055
0.5
0.075
0.05
0.225
0.9

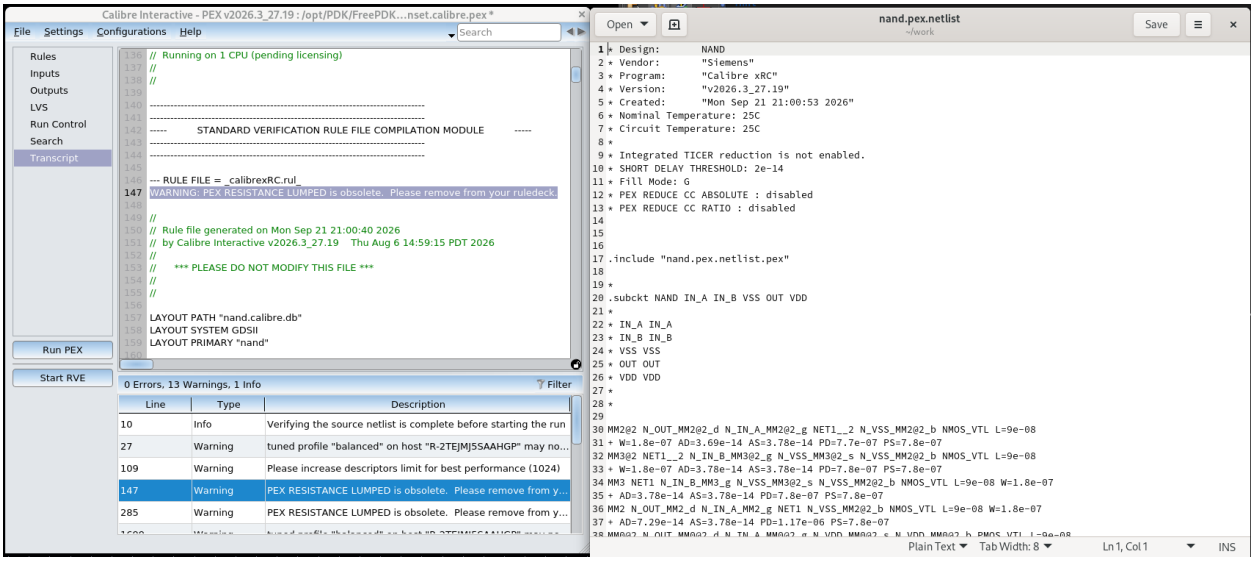
DRC


Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
nand.drc.summary
Run DRC
Show RVE
=== CALIBRE::DRC-H SUMMARY REPORT
Execution Date/Time: Mon Sep 21 20:20:54 2026
Calibre Version: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
Rule File Pathname: _calibreDRC.rul_
Rule File Title:
Layout System: GDS
Layout Path(s): nand.calibre.db
Layout Primary Cell: nand
Current Directory: /home/xuh16/work
User Name: xuh16
Maximum Results/RuleCheck: 1000
Maximum Result Vertices: 4096
DRC Results Database: nand.drc.results (ASCII)
Layout Depth: ALL
Text Depth: PRIMARY
Summary Report File: nand.drc.summary (REPLACE)
Geometry Flagging: ACUTE = NO SKEW = NO ANGLED = NO OFFGRID = NO
NONSIMPLE POLYGON = NO NONSIMPLE PATH = NO
Excluded Cells:
CheckText Mapping: COMMENT TEXT + RULE FILE INFORMATION
Layers: MEMORY-BASED
Keep Empty Checks: YES
--- RUNTIME WARNINGS
--- ORIGINAL LAYER STATISTICS
LAYER pwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER nwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER active ..... TOTAL Original Geometry Count = 4 (4)
LAYER poly ..... TOTAL Original Geometry Count = 9 (9)
LAYER pimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER nimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER vth ..... TOTAL Original Geometry Count = 0 (0)
LAYER vtg ..... TOTAL Original Geometry Count = 0 (0)
LAYER metal1 ..... TOTAL Original Geometry Count = 27 (27)
LAYER metal2 ..... TOTAL Original Geometry Count = 0 (0)
Calibre - RVE v2026.3_27.19 : nand.drc.results
Rule File Pathname: _calibreDRC.rul_
Nwell and Pwell must not overlap
Calibre Run Completed Successfully -- Results are Valid

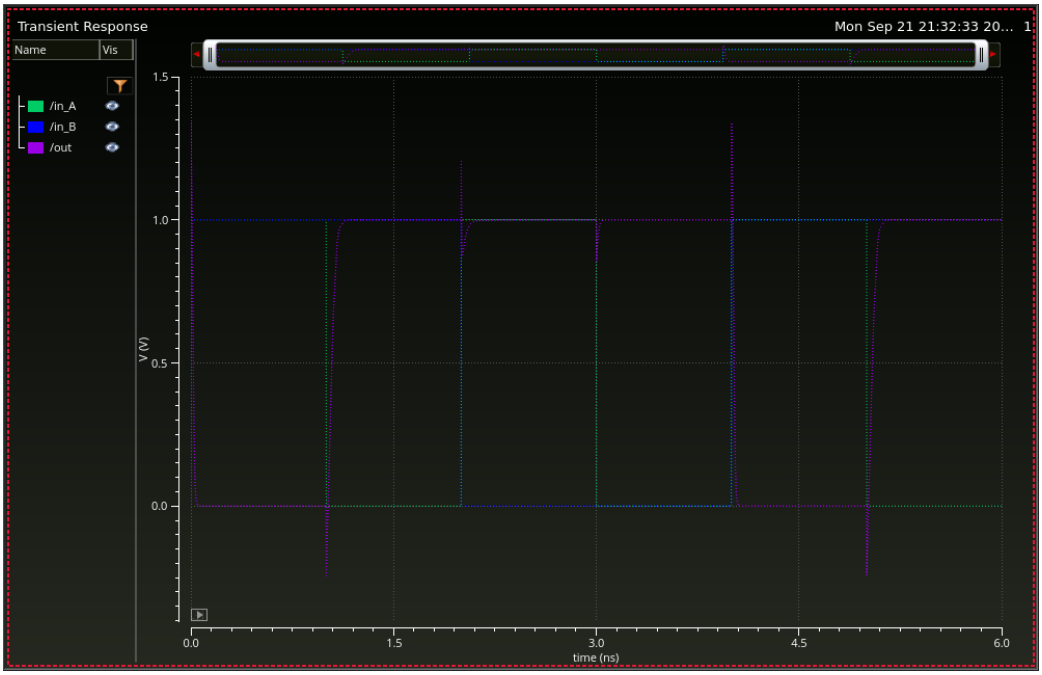
LVS


Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK45/.../runset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
nand.lvs.report
REPORT FILE NAME: nand.lvs.report
LAYOUT NAME: nand.sp ('nand')
SOURCE NAME: nand.src.net ('nand')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Sep 21 13:50:52 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
CELL SUMMARY
Calibre - RVE v2026.3_27.19 : svdb nand
Comparison Results
Cell nand Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
Calibre Run Completed Successfully -- Results are Valid

PEX


Calibre Interactive - PEX v2026.3_27.19 : /opt/PDK/FreePDK...nset.calibre.pex *
File Settings Configurations Help
Rules
Inputs
Outputs
LVS
Run Control
Search
Transcript
Run PEX
Start RVE
// Running on 1 CPU (pending licensing)
STANDARD VERIFICATION RULE FILE COMPILATION MODULE
--- RULE FILE = _calibrexRC.rul_
WARNING: PEX RESISTANCE LUMPED is obsolete. Please remove from your ruledeck.
// Rule file generated on Mon Sep 21 21:00:40 2026
// by Calibre Interactive v2026.3_27.19 Thu Aug 6 14:59:15 PDT 2026
*** PLEASE DO NOT MODIFY THIS FILE ***
LAYOUT PATH "nand.calibre.db"
LAYOUT SYSTEM GDSII
LAYOUT PRIMARY "nand"
0 Errors, 13 Warnings, 1 Info
Filter
Line Type Description
10 Info Verifying the source netlist is complete before starting the run
27 Warning tuned profile "balanced" on host "R-2TEJMJ5SAAHGP" may no...
109 Warning Please increase descriptors limit for best performance (1024)
147 Warning PEX RESISTANCE LUMPED is obsolete. Please remove from y...
285 Warning PEX RESISTANCE LUMPED is obsolete. Please remove from y...
nand.pex.netlist
~/work
Open
Save
* Design: NAND
* Vendor: "Siemens"
* Program: "Calibre xRC"
* Version: "v2026.3_27.19"
* Created: "Mon Sep 21 21:00:53 2026"
* Nominal Temperature: 25C
* Circuit Temperature: 25C
* Integrated TICER reduction is not enabled.
* SHORT DELAY THRESHOLD: 2e-14
* Fill Mode: G
* PEX REDUCE CC ABSOLUTE : disabled
* PEX REDUCE CC RATIO : disabled
.include "nand.pex.netlist.pex"
.subckt NAND IN_A IN_B VSS OUT VDD
* IN_A IN_A
* IN_B IN_B
* VSS VSS
* OUT OUT
* VDD VDD
MM2@2 N_OUT_MM2@2_d N_IN_A_MM2@2_g NET1__2 N_VSS_MM2@2_b NMOS_VTL L=9e-08
+ W=1.8e-07 AD=3.69e-14 AS=3.78e-14 PD=7.7e-07 PS=7.8e-07
MM3@2 NET1__2 N_IN_B_MM3@2_g N_VSS_MM3@2_s N_VSS_MM2@2_b NMOS_VTL L=9e-08
+ W=1.8e-07 AD=3.78e-14 AS=3.78e-14 PD=7.8e-07 PS=7.8e-07
MM3 NET1 N_IN_B_MM3_g N_VSS_MM3@2_s N_VSS_MM2@2_b NMOS_VTL L=9e-08 W=1.8e-07
+ AD=3.78e-14 AS=3.78e-14 PD=7.8e-07 PS=7.8e-07
MM2 N_OUT_MM2_d N_IN_A_MM2_g NET1 N_VSS_MM2@2_b NMOS_VTL L=9e-08 W=1.8e-07
+ AD=7.29e-14 AS=3.78e-14 PD=1.17e-06 PS=7.8e-07
Plain Text
Tab Width: 8
Ln 1, Col 1
INS

Post-Layout Simulation




```
xuh16@R-2TEJMJ5SAAHGP:~/simulation/hw1/nand_simulation/maestro/results/maestro/.tmpADEDir_xuh16/hw1_nand_simulati...
xuh16@R-2TEJMJ5SAAHGP:~/work  ×  xuh16@R-2TEJMJ5SAAHGP:~/simulation/...  ×  xuh16@R-2TEJMJ5SAAHGP:~/simulation/...  ×

****** job statistics summary tnom=  25.000 temp=  25.000 ******

******  Machine Information  ******
CPU:
model name      : Intel(R) Xeon(R) 6975P-C
cpu MHz         : 2700.000
CPU(s)          : 2

OS:
Linux version 5.14.0-687.48.1.el9_8.x86_64 (mockbuild@x86-64-01.build.eng.rdu2.redhat.com) (gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-1
4), GNU ld version 2.35.2-72.el9) #1 SMP PREEMPT_DYNAMIC Sun Sep 13 21:22:30 EDT 2026

System loadavg : 0.24 0.22 0.23 3/688 593717


******  PrimeSim HSPICE Threads Information  ******

Command Line Threads Count :     1
Available CPU Count        :     2
Actual Threads Count       :     1


******  Circuit Statistics  ******
# nodes       =     117 # elements   =     226
# resistors   =      78 # capacitors =     137 # inductors   =       0
# mutual_inds =       0 # vccs       =       0 # vcvs        =       0
# cccs        =       0 # ccvs       =       0 # volt_srcs   =       3
# curr_srcs   =       0 # diodes     =       0 # bjts        =       0
# jfets       =       0 # mosfets    =       8 # U elements  =       0
# T elements  =       0 # W elements =       0 # B elements  =       0
# S elements  =       0 # P elements =       0 # va device   =       0
# vector_srcs =       0 # N elements =       0


******  Runtime Statistics (seconds)  ******

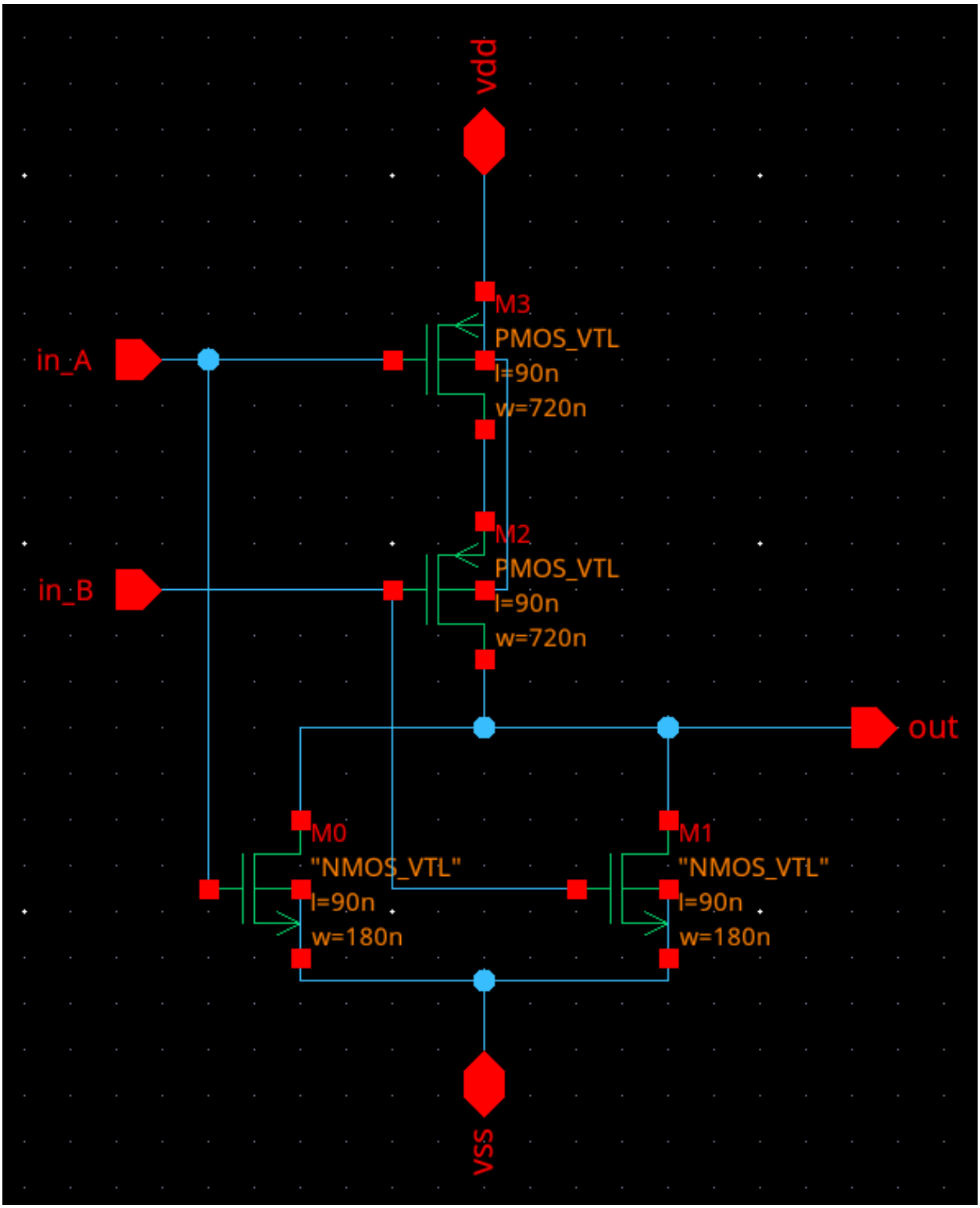
analysis           time    # points   tot. iter  conv.iter
op point           0.00           1          28
transient          0.01         601         385         162 rev=         7
readin             0.01
errchk             0.00
setup              0.00
output             0.00


         peak memory used        473.24 megabytes
         total cpu time            0.02 seconds
         total elapsed time        2.06 seconds
         job started at     21:13:26 09/21/2026
         job ended   at     21:13:28 09/21/2026
         job total runtime         2.06 seconds

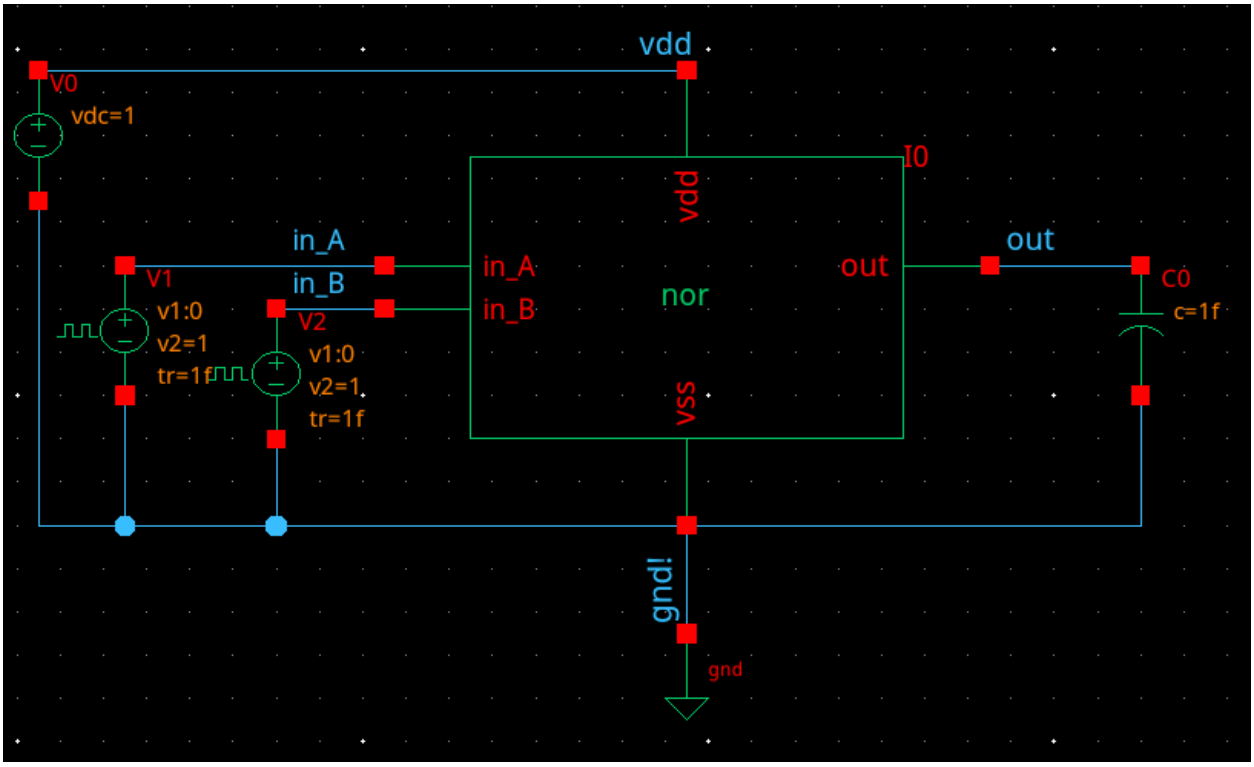

lic: Release hspice token(s)
lic: total license checkout elapse time:        2.03(s)
```

NOR

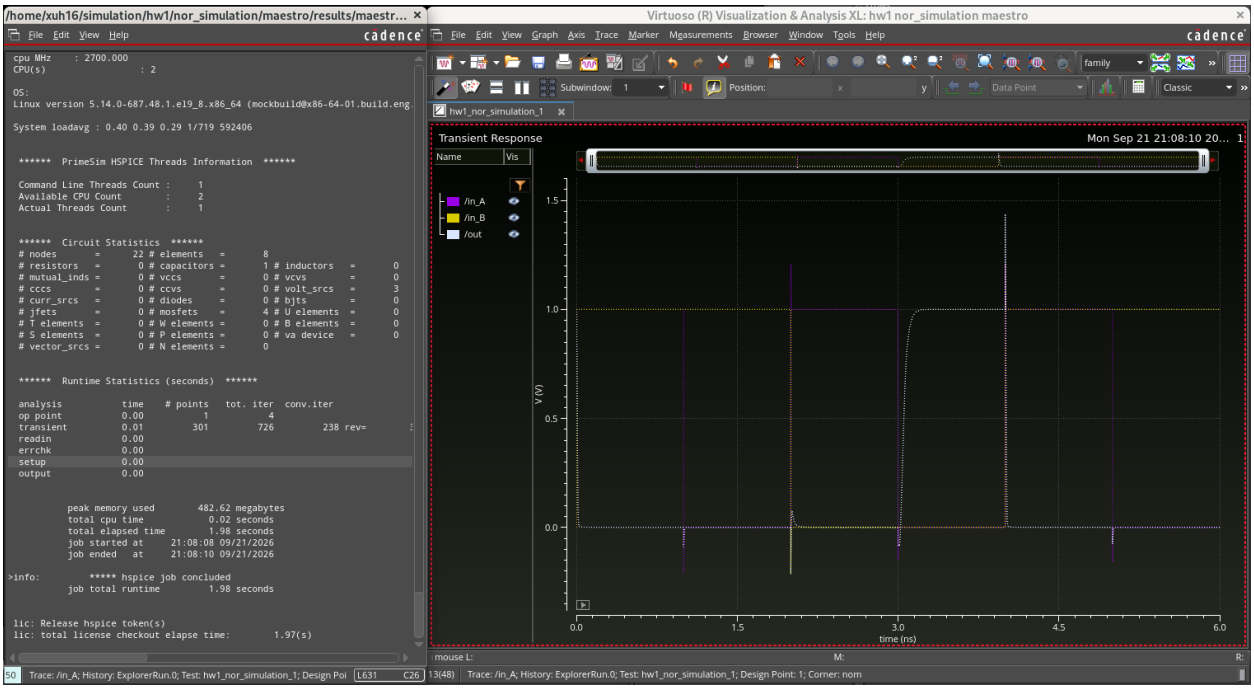
Schematic



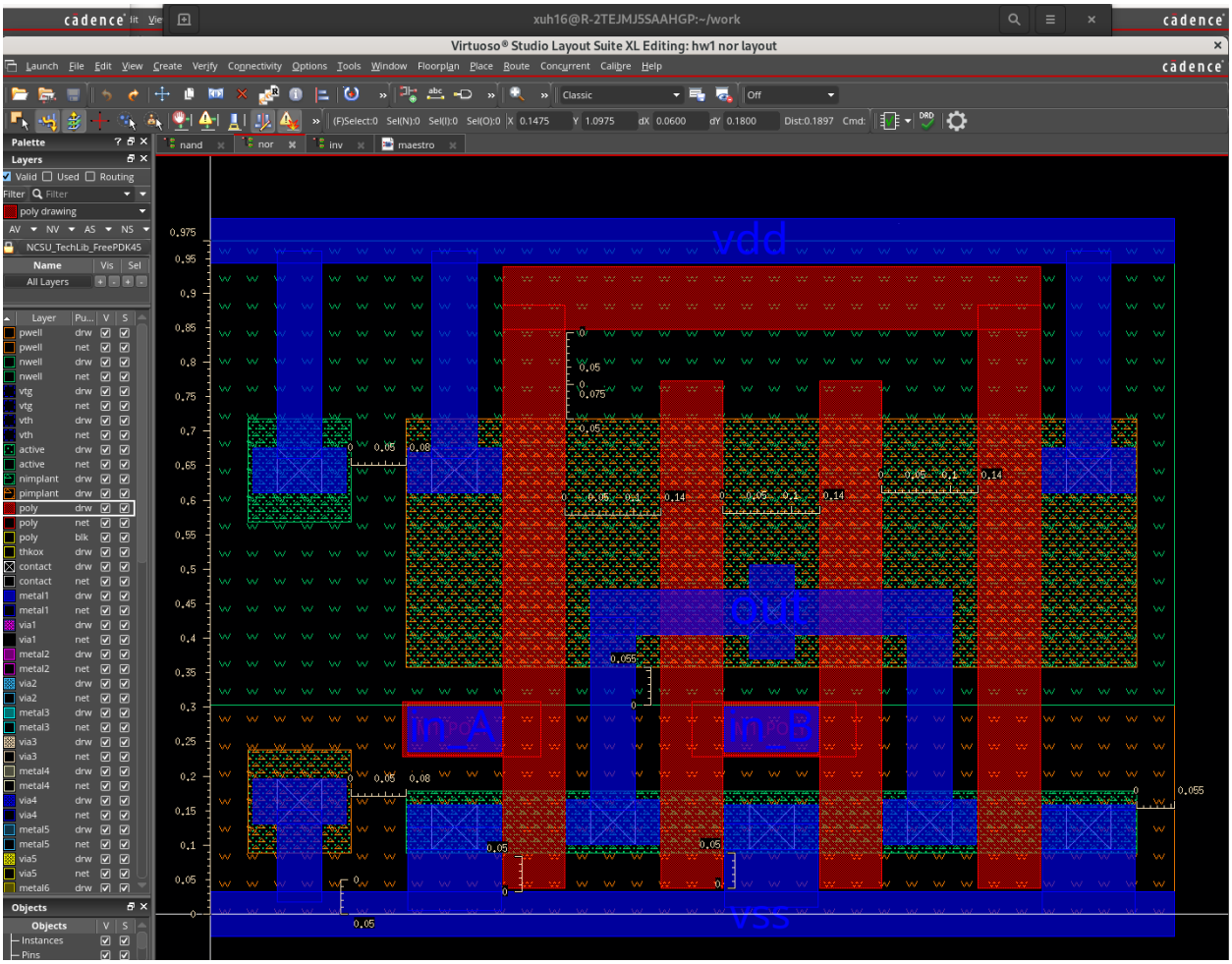
Simulation Schematic



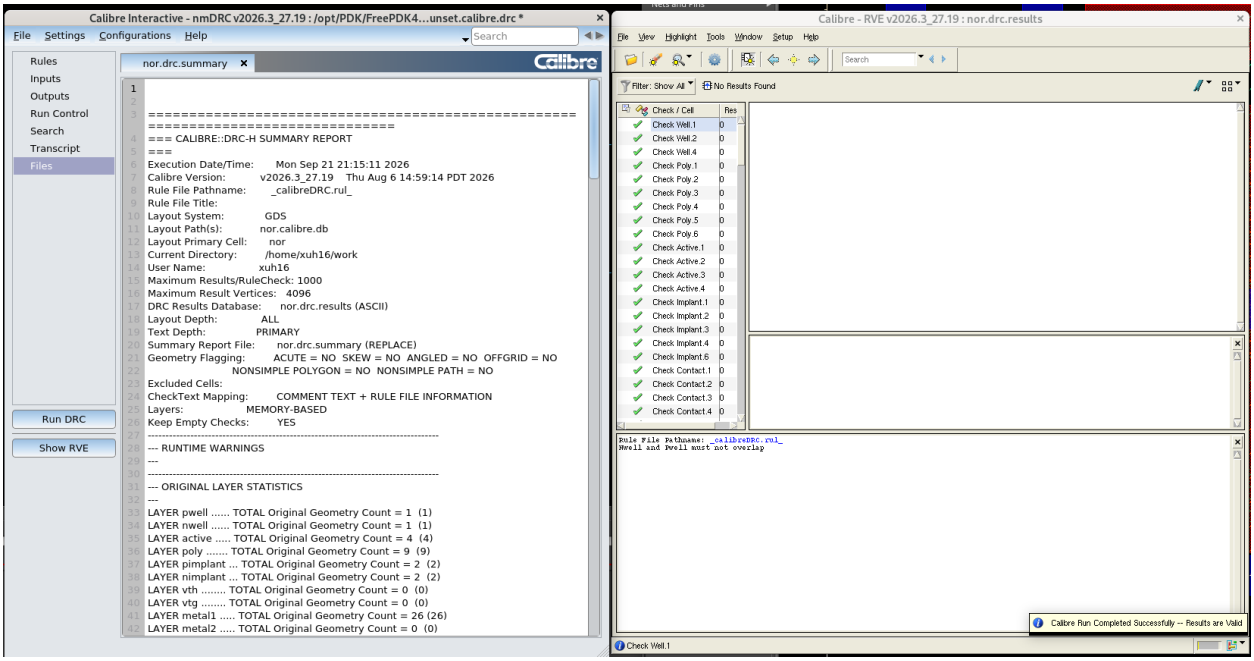
Simulation Results



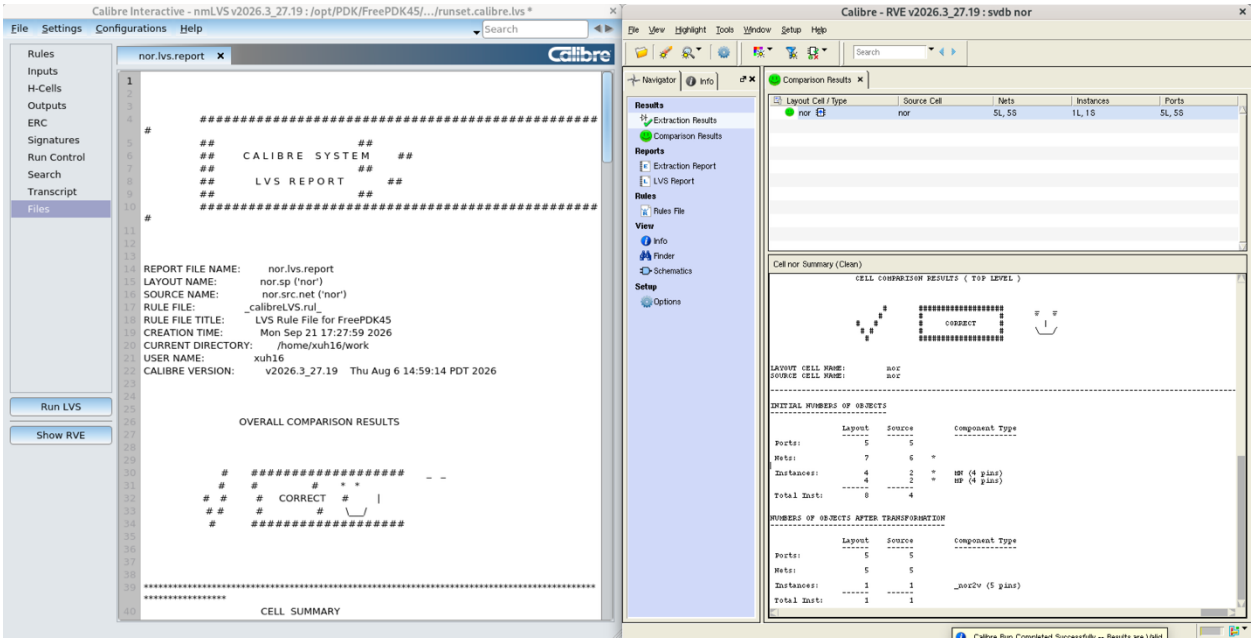
Layout


Virtuoso® Studio Layout Suite XL Editing: hw1 nor layout
vdd
out
inpA
inpB
vss

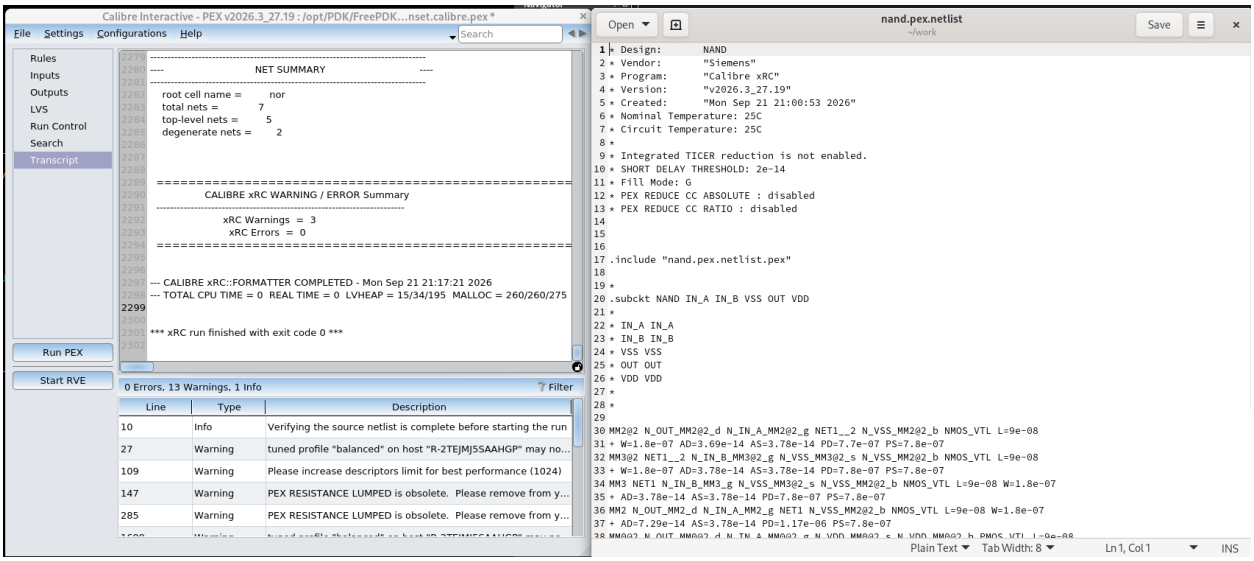
DRC


Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
nor.drc.summary
Calibre - RVE v2026.3_27.19 : nor.drc.results
Run DRC
Show RVE
Calibre Run Completed Successfully -- Results are Valid

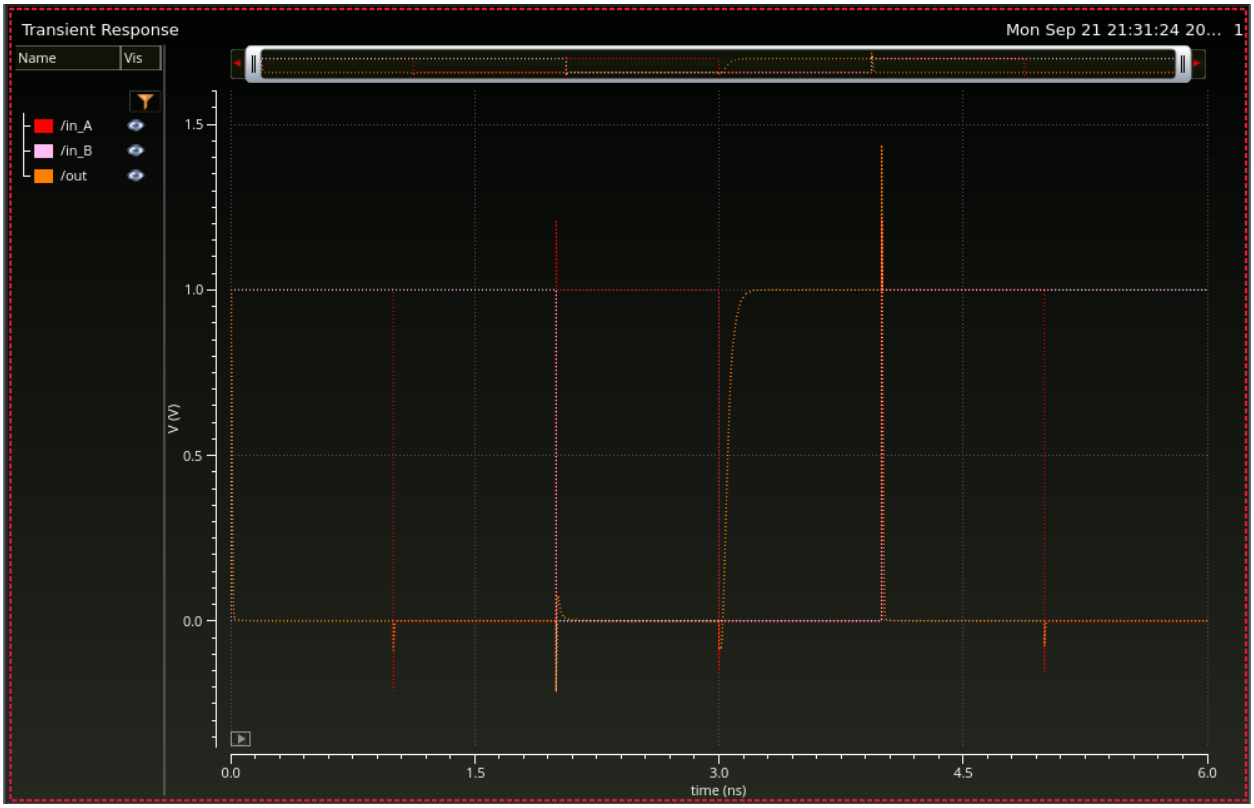
LVS

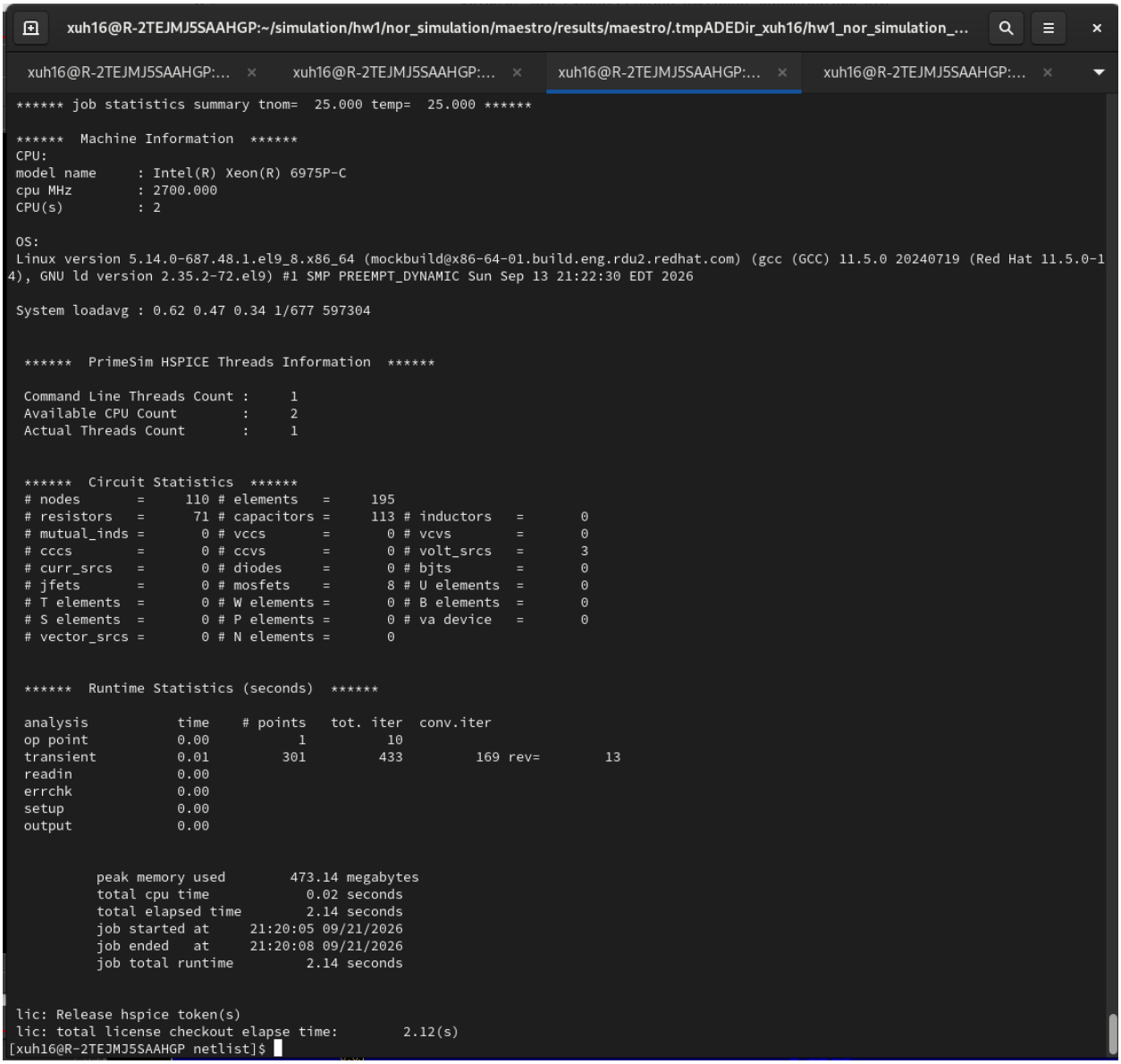

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK45/.../runset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
nor.lvs.report
CALIBRE SYSTEM
LVS REPORT
REPORT FILE NAME: nor.lvs.report
LAYOUT NAME: nor.sp ('nor')
SOURCE NAME: nor.src.net ('nor')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Sep 21 17:27:59 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
CELL SUMMARY
Calibre - RVE v2026.3_27.19 : svdb nor
Navigator
Info
Comparison Results
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Layout Cell / Type
Source Cell
Nets
Instances
Ports
nor
5L, 5S
1L, 1S
Cell nor Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: nor
SOURCE CELL NAME: nor
INITIAL NUMBERS OF OBJECTS
Layout
Source
Component Type
Ports: 5 5
Nets: 7 6 *
Instances: 4 2 * NM (4 pins)
4 2 * MP (4 pins)
Total Inst: 8 4
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Ports: 5 5
Nets: 5 5
Instances: 1 1 _nor2v (5 pins)
Total Inst: 1 1

PEX


Calibre Interactive - PEX v2026.3_27.19 : /opt/PDK/FreePDK...nset.calibre.pex *
File Settings Configurations Help
Rules
Inputs
Outputs
LVS
Run Control
Search
Transcript
Run PEX
Start RVE
NET SUMMARY
root cell name = nor
total nets = 7
top-level nets = 5
degenerate nets = 2
CALIBRE xRC WARNING / ERROR Summary
xRC Warnings = 3
xRC Errors = 0
--- CALIBRE xRC::FORMATTER COMPLETED - Mon Sep 21 21:17:21 2026
--- TOTAL CPU TIME = 0 REAL TIME = 0 LVHEAP = 15/34/195 MALLOC = 260/260/275
*** xRC run finished with exit code 0 ***
0 Errors, 13 Warnings, 1 Info
Filter
Line
Type
Description
10 Info Verifying the source netlist is complete before starting the run
27 Warning tuned profile "balanced" on host "R-2TEJMJ5SAAHGP" may no...
109 Warning Please increase descriptors limit for best performance (1024)
147 Warning PEX RESISTANCE LUMPED is obsolete. Please remove from y...
285 Warning PEX RESISTANCE LUMPED is obsolete. Please remove from y...
Open
nand.pex.netlist
~/work
Save
* Design: NAND
* Vendor: "Siemens"
* Program: "Calibre xRC"
* Version: "v2026.3_27.19"
* Created: "Mon Sep 21 21:00:53 2026"
* Nominal Temperature: 25C
* Circuit Temperature: 25C
* Integrated TICER reduction is not enabled.
* SHORT DELAY THRESHOLD: 2e-14
* Fill Mode: G
* PEX REDUCE CC ABSOLUTE : disabled
* PEX REDUCE CC RATIO : disabled
.include "nand.pex.netlist.pex"
.subckt NAND IN_A IN_B VSS OUT VDD
* IN_A IN_A
* IN_B IN_B
* VSS VSS
* OUT OUT
* VDD VDD
MM2@2 N_OUT_MM2@2_d N_IN_A_MM2@2_g NET1__2 N_VSS_MM2@2_b NMOS_VTL L=9e-08
+ W=1.8e-07 AD=3.69e-14 AS=3.78e-14 PD=7.7e-07 PS=7.8e-07
MM3@2 NET1__2 N_IN_B_MM3@2_g N_VSS_MM3@2_s N_VSS_MM2@2_b NMOS_VTL L=9e-08
+ W=1.8e-07 AD=3.78e-14 AS=3.78e-14 PD=7.8e-07 PS=7.8e-07
MM3 NET1 N_IN_B_MM3_g N_VSS_MM3@2_s N_VSS_MM2@2_b NMOS_VTL L=9e-08 W=1.8e-07
+ AD=3.78e-14 AS=3.78e-14 PD=7.8e-07 PS=7.8e-07
MM2 N_OUT_MM2_d N_IN_A_MM2_g NET1 N_VSS_MM2@2_b NMOS_VTL L=9e-08 W=1.8e-07
+ AD=7.29e-14 AS=3.78e-14 PD=1.17e-06 PS=7.8e-07
Plain Text
Tab Width: 8
Ln 1, Col 1
INS

Post-Layout Simulation




xuh16@R-2TEJMJ5SAAHGP:~/simulation/hw1/nor_simulation/maestro/results/maestro/.tmpADEDir_xuh16/hw1_nor_simulation_...
xuh16@R-2TEJMJ5SAAHGP:...
xuh16@R-2TEJMJ5SAAHGP:...
xuh16@R-2TEJMJ5SAAHGP:...
xuh16@R-2TEJMJ5SAAHGP:...
****** job statistics summary tnom= 25.000 temp= 25.000 ******
****** Machine Information ******
CPU:
model name : Intel(R) Xeon(R) 6975P-C
cpu MHz : 2700.000
CPU(s) : 2
OS:
Linux version 5.14.0-687.48.1.el9_8.x86_64 (mockbuild@x86-64-01.build.eng.rdu2.redhat.com) (gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-14), GNU ld version 2.35.2-72.el9) #1 SMP PREEMPT_DYNAMIC Sun Sep 13 21:22:30 EDT 2026
System loadavg : 0.62 0.47 0.34 1/677 597304
****** PrimeSim HSPICE Threads Information ******
Command Line Threads Count : 1
Available CPU Count : 2
Actual Threads Count : 1
****** Circuit Statistics ******
nodes = 110 # elements = 195
resistors = 71 # capacitors = 113 # inductors = 0
mutual_inds = 0 # vccs = 0 # vcvs = 0
cccs = 0 # ccvs = 0 # volt_srcs = 3
curr_srcs = 0 # diodes = 0 # bjts = 0
jfets = 0 # mosfets = 8 # U elements = 0
T elements = 0 # W elements = 0 # B elements = 0
S elements = 0 # P elements = 0 # va device = 0
vector_srcs = 0 # N elements = 0
****** Runtime Statistics (seconds) ******
analysis time # points tot. iter conv.iter
op point 0.00 1 10
transient 0.01 301 433 169 rev= 13
readin 0.00
errchk 0.00
setup 0.00
output 0.00
peak memory used 473.14 megabytes
total cpu time 0.02 seconds
total elapsed time 2.14 seconds
job started at 21:20:05 09/21/2026
job ended at 21:20:08 09/21/2026
job total runtime 2.14 seconds
lic: Release hspice token(s)
lic: total license checkout elapse time: 2.12(s)
[xuh16@R-2TEJMJ5SAAHGP netlist]$